

XPath Basic Syntax

HTML: <div class="container">...</div>

```
//tagname[@Attribute='Value']
XPath: //div[@class='container']
```

HTML: <h1>Welcome</h1>

Select just a tagname
XPath: //h1

XPath Operators

AND operator

HTML:

```
<input type="radio" name="point" value="1" />
<input type="radio" name="point" value="2" />
<input type="radio" name="point" value="3" />
```

Select second radio button
XPath: //input[@name='point' and @value='2']

OR operator

HTML:

```
<input type="checkbox" class="coding" />
<input type="checkbox" class="music" />
```

Select both checkbox inputs
XPath: //input[@class='coding' or @class='music']

PIPE operator '|' - Combine selectors

HTML:

```
<h1>Welcome to the party</h1>
<h3>Selenium is fun</h3>
<p>XPath is so cool</p>
```

Select all three tags
XPath: //h1 | //h3 | //p

DOT operator - Select element relative to child

HTML:

```
<span>
  <input type="radio" />
</span>
```

Select span which has an input as a direct child
XPath: //span[./input]

Misc

INDEX - Select by index position

HTML:

```
<ul>
  <li>Apple</li>
  <li>Orange</li>
  <li>Kiwi</li>
</ul>
```

Select list item (Orange) with index 2
XPath: //li[2]

WILDCARD - '*' can be substitute with any property or attribute

Select any tagname with class attribute
XPath: //*[@class]

Select tagname div with any attribute
XPath: //div[*]

HTML: <button type="submit" class="btn">...</button>

Select any button with any attribute with value 'btn'
XPath: //button[@*='btn']

XPath Functions

starts-with() - Match in starting attribute value

Dynamic ID part (wt)
HTML: ...

Select starting attribute value
XPath: //a[starts-with(@id, 'u_0_5')]

contains() - Match anywhere in the attribute value

Dynamic session ID (7845263)
HTML: Dashboard

Select anywhere in the node
XPath: //a[contains(@href, 'dashboard/')]

text() - Match exact text content

HTML: <button type="submit">Register</button>

Select button with the text "Register"
XPath: //button[text()='Register']

XPath Axes

PARENT - Selects parent of the current node

HTML:

```
<form name="register">
  <input type="text" name="email" />
</form>
```

Select parent (form) of the current node
XPath: //input[@name='email']/parent::form

CHILD - Selects all child nodes of the current node

HTML:

```
<form name="register">
  <input type="text" name="first-name" />
  <input type="text" name="middle-name" />
  <input type="text" name="last-name" />
  <input type="text" name="email" />
</form>
```

Select all child nodes with tagname input
XPath: //form[@name='register']/child::input

SELF - Selects the current node itself

HTML:

```
<form name="register">
  <input type="text" name="email" />
</form>
```

XPath: //input[@name='email']/self::input

DESCENDANT - Selects all child, grandchild, etc., of current node
(DESCENDANT-OR-SELF - is similar to descendant, but includes self)

HTML:

```
<div name="container">
  <div name="feedback">
    <div name="comment">...</div>
  </div>
</div>
```

Select all descendant nodes with tagname div
XPath: //div[@name='container']/descendant::div

ANCESTOR - Selects all parent, grandparent, etc., of current node
(ANCESTOR-OR-SELF - is similar to ancestor, but includes self)

HTML:

```
<div name="container">
  <div name="feedback">
    <div name="comment">...</div>
  </div>
</div>
```

Select all ancestor nodes with tagname div
XPath: //div[@name='comment']/ancestor::div

XPath Axes - Continued

FOLLOWING - Selects all nodes that appear after the current node

HTML:

```
<div class="profile">
  <input type="text" name="first-name" />
  <input type="text" name="mid-name" />
  <input type="text" name="last-name" />
</div>
<div class="info">
  <input type="text" name="address" />
  <input type="text" name="city" />
</div>
```

Select following nodes with tagname input (last-name, address, city)
XPath: //input[@name='mid-name']/following::input

FOLLOWING-SIBLING - Selects all sibling nodes that appear after the current node

HTML:

```
<div class="profile">
  <input name="first-name" />
  <input name="mid-name" />
  <input name="last-name" />
  <a href="#">...</a>
</div>
```

Select ONLY following sibling nodes which has tagname input (mid-name, last-name)
XPath: //input[@name='first-name']/following-sibling::input

PRECEDING - Selects all nodes that appear before current node

HTML:

```
<div class="profile">
  <input name="first-name" />
  <input name="last-name" />
</div>
<div class="info">
  <input name="address" />
  <input name="city" />
</div>
```

Select preceding nodes with tagname input (first-name, last-name, address)
XPath: //input[@name='city']/preceding::input

PRECEDING-SIBLING - Selects all sibling nodes that appear before current node

HTML:

```
<div class="profile">
  <input name="full-name" />
</div>
<div class="info">
  <input name="address" />
  <input name="city" />
  <input name="state" />
</div>
```

Select ONLY preceding sibling nodes with tagname input (address, city)
XPath: //input[@name='state']/preceding-sibling::input